

BOKU Power BI Migration

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Problem Framing

- ▶ Oracle Warehouse Builder (ETL)
- ▶ SAP BusinessObjects (Reporting)
- ⇒ 20+ years old architecture

- ▶ Fragmented ETL and reporting layers
- ▶ Limited scalability and flexibility

**Replace with a
modern, scalable,
unified data platform**

Oracle WB + SAP BO



Microsoft Power BI

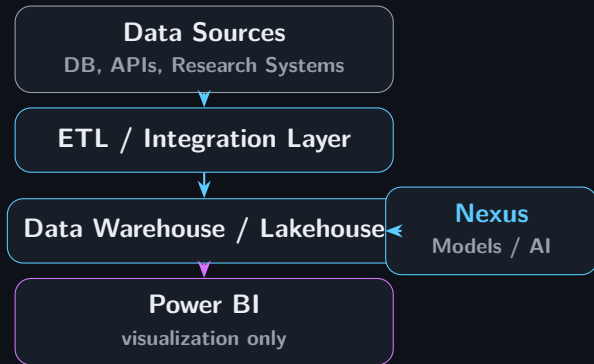
Target Architecture

Key Principle

Clear separation of layers:

1. **Ingestion** — databases, APIs, systems
2. **Storage** — data warehouse / lakehouse
3. **Visualization** — Microsoft Power BI

Power BI = visualization layer
not the data warehouse



Migration Strategy

- ▶ Catalogue all reports and ETL jobs
- ▶ Classify: critical / replaceable / obsolete
- ▶ Map data sources and owners

- ▶ New system runs alongside legacy
- ▶ Pilot reports with key users
- ▶ Validate outputs against old system

No big bang.

- ▶ Go-live per report cluster
- ▶ Rollback plan on standby
- ▶ User training and onboarding
- ▶ Legacy decommission (3 months buffer)

Risks & Pitfalls

Technical Risks

- ▶ Hidden dependencies in ETL processes
- ▶ Data inconsistencies and quality issues
- ▶ Semantic equivalence (same KPI, different calc.)
- ▶ Performance differences after migration

Organizational Risks

- ▶ User resistance (20 years of habit)
- ▶ Lack of training
- ▶ Misaligned stakeholder expectations

- ▶ Automated validation pipelines — old vs. new value-by-value
- ▶ Stakeholder onboarding — key users in pilot group
- ▶ Versioned datasets — no silent overwrites
- ▶ Monitoring and logging — drift detection post-cutover
- ▶ Report triage — usage stats from SAP BO; drop all <5 views/month

Role of Power BI

Strengths

- ▶ Fast dashboard creation
- ▶ Interactive analytics
- ▶ Deep Microsoft ecosystem integration
- ▶ Self-service BI for business users

Technical Capabilities

- ▶ Import Mode — in-memory speed
- ▶ DirectQuery — live source connection
- ▶ Semantic data modelling layer

**Power BI is a
visualization layer**

— not the data warehouse —

Logic, transformations, and
the data model live upstream.

Nexus Layer — Differentiator

Classical BI

Tables → Dashboards

One domain, one tool, one query layer.

Data → **Typed Scientific Objects**

- ▶ Time domain (kinetics, SPR)
- ▶ Reciprocal space (SAXS, $I(q)$)
- ▶ Real space ($p(r)$, geometry)

- ▶ PCA across domains
- ▶ Feature alignment
- ▶ Model-based interpretation

Transition: Reporting →
Computational Science Platform

Final Vision

- ▶ Static reporting system
- ▶ Isolated SAP BO dashboards
- ▶ Manual ETL maintenance
- ▶ No AI/ML readiness

- ▶ Dynamic, scalable data platform
- ▶ AI-ready infrastructure
- ▶ Integrated scientific + admin analytics
- ▶ Foundation for Nexus modeling

Data Warehouse = Foundation for Nexus Faster decisions · Scalable
architecture · Advanced modelling